5. Suppose the inter-industry flow of the product of two industries are given as under.

Production sector	Consumption sector		Domestic demand	Total output
	X	Y		
X	30	40	50	120
Y	20	10	30	60

Determine the technology matrix and test Hawkin's -Simon conditions for the viability of the system. If the domestic demand changes to 80 and 40 units respectively, what should be the gross output of each sector in order to meet the new demands.

6. You are given the following transaction matrix for a two sector economy.

Sector	Sales		Final demand	Gross output
	1	2		
1	4	3	13	20
2	5	4	3	12

(i) Write the technology matrix

(ii) Determine the output when the final demand for the output sector 1 alone increases to 23 units.

7. Suppose the inter-industry flow of the product of two sectors X and Y are given as under.

Production sector	Consumption Sector		Domestic demand	Gross output
	X	Y		
X	15	10	10	35
Y	20	30	15	65

Find the gross output when the domestic demand changes to 12 for X and 18 for Y.

Exercise 1.5

Choose the correct answer

1. The value of x if $\begin{vmatrix} 0 & 1 & 0 \\ x & 2 & x \\ 1 & 3 & x \end{vmatrix} = 0$ is

(a) 0, –1 (b) 0, 1

(c) –1, 1 (d) –1, –1

2. The value of $\begin{vmatrix} 2x+y & x & y \\ 2y+z & y & z \\ 2z+x & z & x \end{vmatrix}$ is

(a) $x\,y\,z$ (b) $x + y + z$

(c) $2x + 2y + 2z$ (d) 0

3. The cofactor of –7 in the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$ is

I8EZYE

(a) –18 (b) 18

(c) –7 (d) 7

4. If $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{vmatrix}$ then $\begin{vmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{vmatrix}$ is

(a) Δ (b) $-\Delta$

(c) 3Δ (d) -3Δ

5. The value of the determinant $\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix}^2$ is

(a) abc (b) 0

(c) $a^2\,b^2\,c^2$ (d) $-abc$

01_11th_BM-STAT_Ch-1-EM.indd 20 21-04-2020 12:20:12 PM